

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer. Expertise in building scalable microservices and distributed data pipelines, with a Master of Science in Computer Science focusing on large-scale data platforms and privacy-preserving AI. Proficient in Java, Python, Spark, SQL, AWS, and Azure, with experience in CI/CD automation and a strong foundation in React, Hooks, Redux, and TypeScript for full-stack development. Eager to leverage full-stack capabilities in a Software Engineer I, Fullstack, Risk Engineering role.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered CI/CD pipelines via Jenkins and Azure DevOps, automating deployments across Azure Cloud and Oracle WebLogic environments, enhancing release velocity.
- Modernized legacy systems by migrating to Log4j 2.x and implementing robust monitoring, reducing issue resolution time by 20 percent.
- Spearheaded an EDX platform upgrade, streamlining application architecture and dependency management, leading to 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git, SVN, and comprehensive JUnit testing for quality assurance.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for major automotive clients, ensuring seamless integration.
- Improved system performance by 90 percent through strategic SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented streamlined CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for consistent software delivery.
- Mentored junior developers on clean coding practices, code reviews, and effective debugging techniques, fostering team growth.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization strategies.
- Enhanced SQL data pipelines and database operations, achieving a 10 percent improvement in query performance and bolstering data integrity.
- Implemented comprehensive unit testing using JUnit4, consistently reaching 99 percent code coverage for critical application modules.
- Supported release planning and sprint execution within Agile development teams using JIRA, ensuring timely project milestones.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline processing over 8 GB of NYC taxi trip data in near real-time, enabling rapid insights.
- Built an interactive Flask web application with dynamic filtering and visualization for intuitive data exploration and analysis.
- Containerized ingestion, processing, and analytics components using Docker for reproducible and scalable deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, enabling infrastructure as code.

Privacy-Preserving AI Email Client (Capstone Project - Python, Hugging Face, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models using Python, facilitating email summarization and classification tasks.
- Applied parameter-efficient fine-tuning techniques with LoRA to support low-resource training under strict privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment of AI models.

SKILLS

- Programming Languages: Proficient in Java and Python for robust application development and data analysis, complemented by extensive SQL query expertise for complex data manipulation.
- Full-Stack Development: Skilled in modern web technologies including React, Hooks, and Redux, leveraging TypeScript for enhanced code quality and maintainability.
- Distributed Systems & Data Engineering: Experienced with Apache Spark, PySpark for large-scale data processing, and building

resilient ETL pipelines and data warehousing solutions.

- Cloud Platforms & Infrastructure: Adept in AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps), with practical experience in infrastructure automation using Terraform.
- Databases & Tools: Proficient with Relational Databases like PostgreSQL and Oracle, and familiar with CI/CD tools such as Jenkins and Git for efficient workflow management.